

Myocardial Fibroblast Activation in Ischemic and Nonischemic Cardiomyopathy

Shruti S. Joshi, MBBS, PhD; Anna K. Barton, MBChB; Krithika Loganath, MBChB; Neil J. Craig, PhD; Menaka Mahendran, MBBS; Beth Whittington, MBChB; Tariq Ramtoola, MBChB; Michelle C. Williams, MBChB, PhD; Edwin J. R. van Beek, MBChB, PhD; Andrew H. Baker, PhD; Alison Fletcher, PhD; Tim Clark, PhD; Clint Waight, PhD; Damini Dey, PhD; Piotr Slomka, PhD; David E. Newby, MBChB, PhD; Marc R. Dweck, MBChB, PhD

IMPORTANCE Adverse myocardial remodeling and fibrosis contribute to heart failure progression and are thought to be driven by activated fibroblasts. Noninvasive assessment of myocardial fibroblast activation may improve phenotyping and risk stratification in heart failure.

OBJECTIVE To evaluate myocardial fibroblast activation in patients with heart failure with reduced ejection fraction using gallium 68-labeled fibroblast activation protein inhibitor 46 (^{68}Ga]FAPi-46) positron emission tomography (PET) and magnetic resonance imaging (MRI).

DESIGN, SETTING, AND PARTICIPANTS This was a prospective case-control study including patients with heart failure with reduced ejection fraction, patients with prior myocardial infarction without heart failure, and healthy volunteers. A subset of patients with heart failure underwent repeat imaging after more than 6 months. Data analysis was conducted from January 2024 to January 2025.

EXPOSURE All participants underwent ^{68}Ga]FAPi-46 PET/MRI.

MAIN OUTCOMES AND MEASURES Myocardial fibroblast activation quantified using maximum standardized uptake values (SUVmax) of ^{68}Ga]FAPi-46.

RESULTS A total of 81 participants were included (mean [SD] age, 66.2 [9.7] years; 22 [27%] female): 42 with heart failure (21 with heart failure due to ischemic cardiomyopathy from a previous myocardial infarction and 21 with a nonischemic etiology; mean [SD] left ventricular ejection fraction, 41% [9%]), 20 with a prior myocardial infarction but preserved left ventricular systolic function, and 19 healthy volunteers. No myocardial fibroblast activation was observed in healthy volunteers. All patients with heart failure demonstrated increased myocardial ^{68}Ga]FAPi-46 uptake compared with healthy volunteers (mean [SD] SUVmax, 2.7 [1.5] vs 1.5 [0.3]; $P < .001$). Uptake was highest in patients with ischemic cardiomyopathy, localizing to regions of established myocardial infarction (mean [SD] SUVmax, 3.2 [1.1]). Patients with nonischemic cardiomyopathy exhibited a different pattern of diffuse, lower-intensity uptake (mean [SD] SUVmax, 2.3 [0.5]), with the highest signal in the basal septum irrespective of late gadolinium enhancement. Patients with ischemic cardiomyopathy had higher uptake than patients with prior myocardial infarction without heart failure ($n = 20$) despite no major difference in infarct size (mean [SD] SUVmax, 3.2 [1.1] vs 2.5 [0.3]; $P = .03$). Among patients with heart failure who underwent repeat imaging, higher baseline ^{68}Ga]FAPi-46 uptake was associated with less improvement in ejection fraction with optimal medical therapy over time ($r = -0.52$; $P = .02$).

CONCLUSIONS AND RELEVANCE In this case-control study, patients with heart failure demonstrated persistent myocardial fibroblast activation with distinct spatial patterns according to cardiomyopathy etiology. Noninvasive imaging of fibroblast activation may provide mechanistic insights, aid risk stratification, and support the development of targeted antifibrotic therapies in heart failure.

JAMA Cardiol. doi:10.1001/jamacardio.2026.0661
Published online April 15, 2026.

- [+ Editor's Note](#)
- [+ Related article](#)
- [+ Supplemental content](#)

Author Affiliations: British Heart Foundation Centre of Research Excellence, University of Edinburgh, Edinburgh, United Kingdom (Joshi, Barton, Loganath, Craig, Mahendran, Whittington, Ramtoola, Williams, van Beek, Baker, Fletcher, Clark, Waight, Newby, Dweck); Edinburgh Heart Centre, Royal Infirmary of Edinburgh, Edinburgh, United Kingdom (Joshi, Barton, Williams, van Beek, Waight, Newby, Dweck); Biomedical Imaging Research Institute, Cedars-Sinai Medical Center, Los Angeles, California (Dey, Slomka).

Corresponding Author: Shruti S. Joshi, MBBS, PhD, British Heart Foundation Centre of Research Excellence, University of Edinburgh, 49 Little France Crescent, Edinburgh EH16 4TJ, United Kingdom (sjoshi@ed.ac.uk).

Heat failure remains one of the leading causes of death and disability worldwide, and despite major improvements in treatment strategies, survival rates after a diagnosis of heart failure continue to be poor.¹ Left ventricular (LV) remodeling is a key mechanism in the progression of heart failure and is characterized by changes in volumes, geometry, and dimensions as well as impairment in systolic and diastolic function. At the cellular and tissue levels, this remodeling is driven by a combination of pathologic myocyte hypertrophy, apoptosis, activated fibroblast proliferation, and myocardial fibrosis.²

Activated fibroblasts are the main effector cells driving fibrosis and are believed to be key mediators of adverse remodeling of the myocardium and the development of heart failure. They are not present until quiescent fibroblasts are activated by pathologic triggers, such as myocardial infarction, pressure overload, and myocardial injury.³ Their activation is closely associated with the activity of several pathways, including the renin-angiotensin-aldosterone system (RAAS) and inflammatory cytokines.^{4,5}

The imaging of activated fibroblasts in the heart is now feasible with positron emission tomography (PET) using the radiotracer gallium 68-labeled fibroblast activation protein (FAP) inhibitor 46 (⁶⁸Ga)FAPi-46.^{6,7} This tracer is highly specific for FAP, which is expressed exclusively on the surface of activated fibroblasts. [⁶⁸Ga]FAPi-46 can detect activated fibroblasts in patients with cancer as well as patients with acute myocardial infarction.⁶⁻⁹ In the latter, it provides distinct but complementary information to the presence of myocardial fibrosis detected by cardiac magnetic resonance imaging (MRI) using late gadolinium enhancement and T1 mapping techniques.¹⁰⁻¹² Furthermore, multiple preclinical and early clinical studies have explored the use of chimeric antigen receptor (CAR) T-cell therapy in cardiovascular diseases, establishing the potential of targeting FAP-positive fibroblasts with this approach as a novel therapeutic strategy.¹³ Consequently, a noninvasive imaging technique able to image these activated fibroblasts directly would prove useful in supporting the development and evaluation of emerging therapies targeting these cells.

In this prospective case-control study, we investigated myocardial fibroblast activation in patients with heart failure using [⁶⁸Ga]FAPi-46 PET and cardiac MRI. We assessed fibroblast activation in patients with heart failure due to ischemic or nonischemic cardiomyopathies and compared it with that in patients with prior myocardial infarction without heart failure and healthy volunteers.

Methods

Study Design

We conducted a prospective case-control study in accordance with the Declaration of Helsinki, with approval from the research ethics committee and the written informed consent of all participants. Data analysis was conducted from January 2024 to January 2025.

Study Participants

Patients with a confirmed diagnosis of heart failure,¹⁴ with an LV ejection fraction less than 50%, and aged 50 years or older

Key Points

Question What are the patterns of myocardial fibroblast activation in patients with different etiologies of heart failure?

Findings In this case-control study, gallium 68-labeled fibroblast activation protein inhibitor 46 ([⁶⁸Ga]FAPi-46) positron emission tomography (PET) and magnetic resonance imaging (MRI) demonstrated increased myocardial fibroblast activation in patients with heart failure; uptake was highest in ischemic cardiomyopathy, localizing to prior infarct, whereas patients with nonischemic cardiomyopathy showed a diffuse, lower-intensity pattern with predominant septal involvement. Fibroblast activation remained stable on repeat imaging, with increased baseline activity associated with less improvement in ejection fraction.

Meaning [⁶⁸Ga]FAPi-46 PET/MRI demonstrated increased myocardial fibroblast activation in patients with heart failure and may represent an important target for disease-modifying therapy.

were identified and approached through the Edinburgh Heart Centre, National Health Service Lothian, Edinburgh, United Kingdom. All study participants had stable heart failure and needed to have been receiving a stable regimen of optimal heart failure medications for 3 months prior to inclusion. These participants were recruited and scanned between August 2021 and August 2023. Patients with heart failure were screened using transthoracic echocardiography prior to their inclusion in the study. These patients were categorized by ischemic or nonischemic etiology based on their clinical history, angiographic findings, and pattern of late gadolinium enhancement.¹⁵ We also recruited a cohort of patients with a history of myocardial infarction but preserved systolic function defined by an ejection fraction greater than 50% and a cohort of healthy volunteers with no history of cardiovascular disease, good LV systolic function (ejection fraction >50%), and no evidence of late gadolinium enhancement on cardiac MRI.

Exclusion criteria for all study participants included kidney impairment (estimated glomerular filtration rate <30 mL/min/1.73 m²), inability to provide consent, contraindications to MRI, pregnancy, or breastfeeding. All participants underwent [⁶⁸Ga]FAPi-46 PET/MRI at baseline, and a subset of patients with heart failure underwent repeat imaging 6 to 12 months following their baseline scan (eFigure 1 in Supplement 1).

PET and Cardiac MRI

Imaging was performed using a hybrid PET/MRI scanner (Biograph mMR; Siemens Healthineers) with a 50-minute bed time, centered over the heart, 30 to 45 minutes following administration of 180 to 200 MBq of [⁶⁸Ga]FAPi-46. The MRI protocol included long-axis cine imaging (2-, 3-, and 4-chamber views) and a short-axis ventricular cine stack. The protocol included late gadolinium enhancement images acquired in 2 full short-axis stacks in 2 phase-encoding directions 7 to 15 minutes after administration of 1-mg/kg gadobutrol (Gadovist). PET images were reconstructed using an ordered subset expectation maximization iterative algorithm, undergoing 3 iterations of 21 subsets. Reconstructions were scaled to a 344 × 344-pixel matrix with a zoom factor of 1, giving voxel

Table 1. Baseline Characteristics of Patients With Heart Failure (HF) vs Healthy Volunteers

Characteristic	Healthy volunteers (n = 19)	Patients with previous MI with preserved systolic function (n = 20)	Patients with HF (n = 42)	P value, patients with HF vs healthy volunteers	Patients with ischemic cardiomyopathy (n = 21)	P value, patients ischemic cardiomyopathy vs patients with previous MI with preserved systolic function	Patients with nonischemic cardiomyopathy (n = 21)	P value, patients with nonischemic vs ischemic cardiomyopathy
Age, mean (SD), y	60 (6)	69 (10)	68 (9)	.01	66 (9)	.21	68 (10)	.87
Sex, No. (%)								
Female	6 (32)	5 (25)	11 (26)	NA	5 (24)	NA	6 (29)	NA
Male	13 (68)	15 (75)	31 (74)	NA	16 (76)	NA	15 (71)	NA
BMI, mean (SD)	27.4 (5.4)	28.2 (4.0)	29.6 (5.0)	.18	28.5 (4.7)	.24	28.8 (4.8)	.43
High-sensitivity cardiac troponin T, mean (SD), ng/L	NA	NA	39 300 (74 600)	NA	60 000 (92 000)	NA	15 000 (9000)	.01
B-type natriuretic peptide, mean (SD), pg/mL	NA	NA	1235 (1716)	NA	1805 (2030)	NA	1143 (1660)	.14
β-Blocker therapy, No. (%)	NA	15 (75)	37 (87)	NA	19 (86)	NA	21 (91)	NA
Angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker therapy, No. (%)	NA	17 (85)	19 (44)	NA	7 (31)	NA	12 (52)	NA
Sacubitril and valsartan, No. (%)	NA	NA	23 (54)	NA	14 (63)	NA	11 (48)	NA
Mineralocorticoid receptor antagonists, No. (%)	NA	NA	26 (62)	NA	15 (69)	NA	13 (57)	NA

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); HF, heart failure; MI, myocardial infarction; NA, not applicable.

SI conversion factors: To convert high-sensitivity cardiac troponin T to micrograms per liter, multiply by 0.001; B-type natriuretic peptide to nanograms per liter, multiply by 1.

dimensions of $2.09 \times 2.09 \times 2.03$ mm. Gaussian smoothing was applied with a 5-mm full-width at half-maximum kernel. Attenuation correction was performed using a free-breathing, 3-dimensional, golden-angle radial, spoiled gradient echo sequence (GRE) acquisition (free-breathing radial GRE) to reduce artifacts.¹⁶

Image Analysis

Image analysis was performed at an imaging core laboratory (Edinburgh Imaging, University of Edinburgh) offline and blinded to all participant details. Cardiac chamber volumes, mass, and function were assessed using cvi42 version 5.13.5 (Circle Cardiovascular Imaging). Endocardial and epicardial borders were defined on the short-axis cine images for volumetric and wall-motion measurements.

PET image analysis was performed using a dedicated image analysis software platform, FusionQuant (Cedars-Sinai Medical Center). This allows for precise image coregistration, automated motion correction, and advanced cardiac PET quantification. We measured maximum uptake in the myocardium (infarct zone or usually the basal septum for patients without ischemia) as well as uptake in the myocardium remote from the maximal signal. We measured [⁶⁸Ga]FAPI-46 uptake in 16 of the 17 myocardial segments, with the true apical segment excluded because it was too thin to assess radiotracer uptake reliably. Maximum standardized uptake value (SUVmax) and maximum target to background ratio

were calculated for all regions of interest, as described previously.¹⁷

Statistical Analysis

Categorical baseline variables were presented as number (percentage) and compared using χ^2 test. Continuous data were assessed for normality using the Shapiro-Wilk test and presented as mean (SD) or median (IQR). Values were compared using analysis of variance and Tukey post hoc test, paired or unpaired *t* tests, Wilcoxon or Mann-Whitney tests, and Kruskal-Wallis tests as appropriate. Stepwise regression model to assess predictors of myocardial fibroblast activation included variables of age, sex, high-sensitivity cardiac troponin T, N-terminal pro-B-type natriuretic peptide (BNP), and treatments with angiotensin receptor blockers, angiotensin-converting enzyme inhibitors, β-blockers, mineralocorticoid receptor antagonists, and angiotensin receptor neprilysin inhibitors. Statistical significance was taken as 2-sided *P* < .05. All statistical analyses were performed with R Studio software version 1.3.1093 (RStudio).

Results

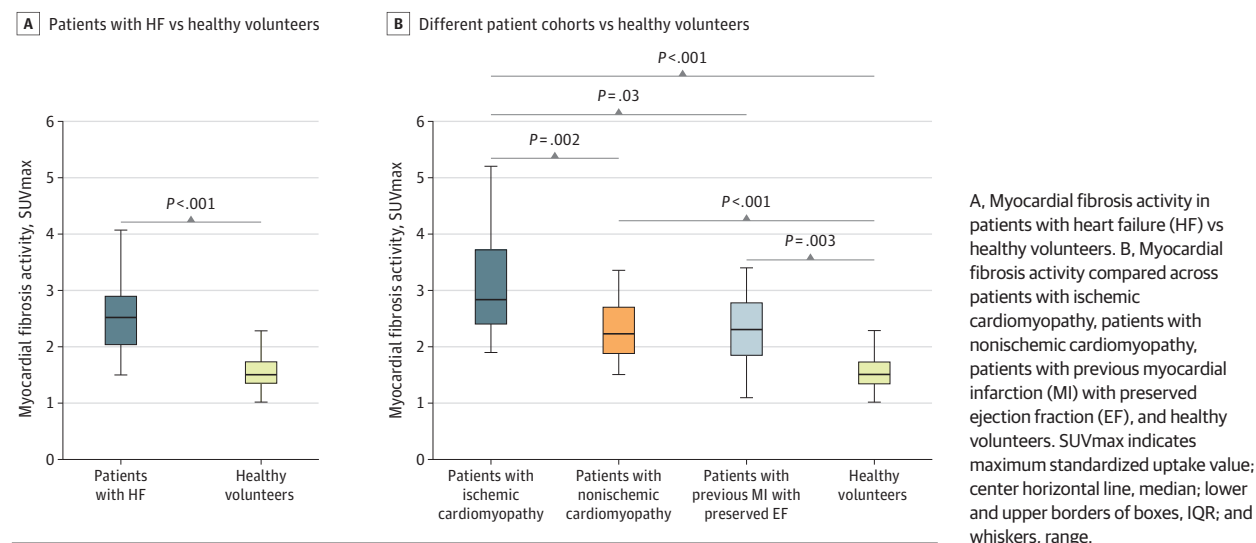
We recruited a total of 81 participants (mean [SD] age, 66.2 [9.7] years; 22 [27%] female): 42 patients with heart failure (21 with heart failure due to ischemic cardiomyopathy from a previous

Table 2. Magnetic Resonance Imaging of Study Population

Finding	Healthy volunteers	Patients with HF	P value, patients with HF vs healthy volunteers	Patients with ischemic cardiomyopathy	Patients with nonischemic cardiomyopathy	P value, patients with nonischemic vs ischemic cardiomyopathy	Patients with previous MI with preserved EF	P value, patients with ischemic cardiomyopathy vs patients with previous MI and preserved EF
Indexed end-diastolic volume, mean (SD), mL/m ²	69.8 (11.0)	102.4 (47.7)	<.001	103.4 (26.7)	102.3 (58.3)	.99	73.0 (12.8)	<.001
Indexed end-systolic volume, mean (SD), mL/m ²	24.0 (6.6)	64.3 (39.0)	<.001	64.7 (24.6)	64.6 (46.5)	.99	28.9 (7.3)	<.001
Indexed LV mass, mean (SD), g/m ²	61.6 (11.5)	85.3 (25.9)	<.001	85.2 (23.5)	85.2 (26.3)	.99	62.3 (12.1)	<.001
LVEF, mean (SD), %	65.8 (5.7)	41.0 (8.6)	<.001	40.1 (9.0)	41.0 (8.2)	.52	60.5 (6.7)	<.001
Late gadolinium enhancement mass, mean (SD), g	NA	9.4 (16.1)	NA	14.4 (21.0)	2.9 (4.5)	<.001	10.3 (8.0)	.11

Abbreviations: EF, ejection fraction; HF, heart failure; LV, left ventricular; LVEF, left ventricular ejection fraction; MI, myocardial infarction; NA, not applicable.

Figure 1. Box-and-Whisker Plots of Myocardial Fibrosis Activity



myocardial infarction and 21 patients with a nonischemic etiology), 20 patients with a prior myocardial infarction but preserved LV systolic function, and 19 healthy volunteers. Participants were predominantly older men who were generally well matched between groups (Table 1). Patients with heart failure were well treated in terms of guideline-directed medical therapy¹⁸ over the period of recruitment and follow-up, which preceded routine use of sodium-glucose cotransporter-2 inhibitor therapy (Table 1). Administration of [⁶⁸Ga]FAP-46 and PET/MRI were well tolerated in all participants with no adverse reactions.

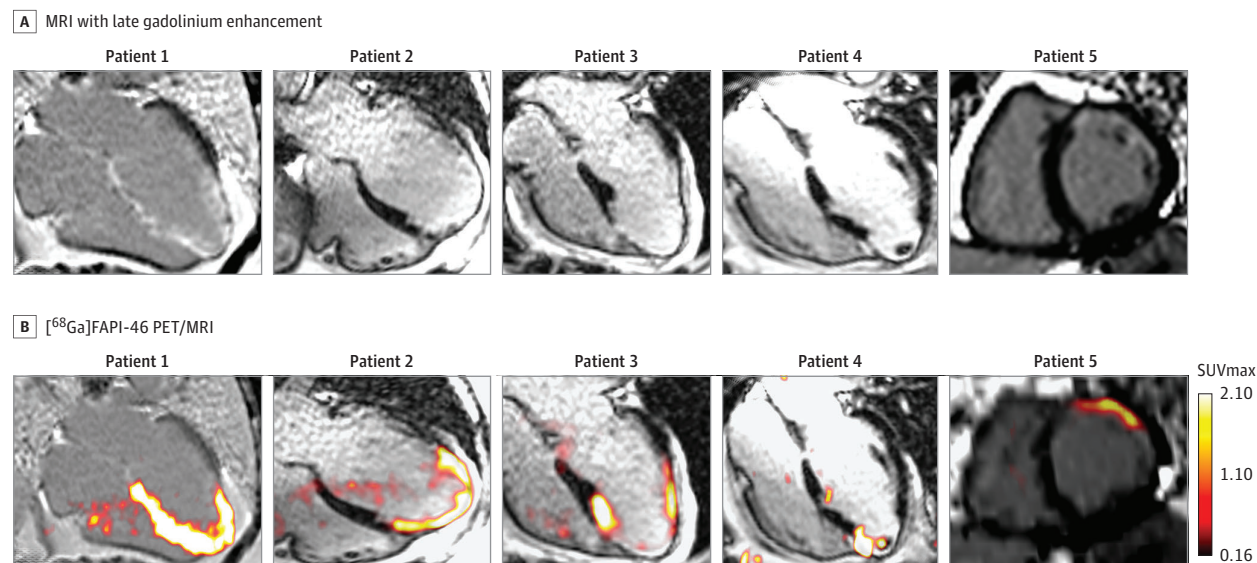
Patients with heart failure had reduced LV ejection fraction (mean [SD], 41% [9%]) and a high burden of myocardial fibrosis. All patients with ischemic cardiomyopathy were scanned at least 2 years after myocardial infarction, with a mean (SD) infarct-to-imaging interval of 8.2 (8.0) years. These 21 patients with ischemic cardiomyopathy demonstrated evidence of infarct-pattern late gadolinium enhancement: 12 (57%) had an anterior infarct and 9 (43%) had either an

inferior or inferolateral infarct. By contrast, 10 of the 21 patients with nonischemic cardiomyopathy demonstrated non-infarct-pattern late gadolinium enhancement. These patients with nonischemic cardiomyopathy comprised 16 patients with dilated cardiomyopathy, 2 with hypertensive heart disease, and 3 with diabetic cardiomyopathy. Healthy volunteers had a normal LV ejection fraction and no evidence of late gadolinium enhancement (Table 2).

Patients With Heart Failure

Myocardial [⁶⁸Ga]FAP-46 uptake was higher in patients with heart failure and reduced ejection fraction compared with healthy volunteers (mean [SD] SUVmax, 2.7 [1.5] vs 1.5 [0.3]; $P < .001$) (Figure 1). The 2 cohorts of ischemic and nonischemic cardiomyopathy were well matched for LV ejection fraction and indexed end-diastolic or end-systolic volumes (Table 2). However, patients with ischemic cardiomyopathy had higher scar burden measured by the mass of late gadolinium enhancement com-

Figure 2. Examples of Patients With Previous Myocardial Infarction



A, Magnetic resonance imaging (MRI) with late gadolinium enhancement. B, Fused MRI and positron emission tomographic (PET) images. Patients 1 through 3 had ischemic cardiomyopathy (previous myocardial infarction with reduced ejection fraction). For patients 1 and 2, the highest myocardial fibrosis activity is seen in the scar region, as shown in the images with late gadolinium enhancement. For patient 3, the highest myocardial fibrosis activity is seen in the scar hinge points, as opposed to the entire scar in contrast to patients 1 and

2. Patients 4 and 5 had previous myocardial infarction with preserved ejection fraction. Areas of fibroblast activation are still observed using gallium 68-labeled fibroblast protein inhibitor 46 ([^{68}Ga]FAPI-46), but these are of lower intensity and less extensive than in patients with previous myocardial infarction and reduced ejection fraction. Patient 4 demonstrates evidence of a left ventricular thrombus that also shows increased [^{68}Ga]FAPI-46 uptake. SUVmax indicates maximum standardized uptake value.

pared with those with heart failure due to nonischemic cardiomyopathy (mean [SD], 14.4 [21.0] vs 2.9 [4.5] g; $P < .001$).

Myocardial [^{68}Ga]FAPI-46 uptake was higher in the patients with ischemic cardiomyopathy compared with those with a nonischemic etiology (mean [SD] SUVmax, 3.2 [1.1] vs 2.3 [0.5]; $P = .002$). However, those with nonischemic cardiomyopathy still demonstrated higher myocardial [^{68}Ga]FAPI-46 uptake compared with the healthy volunteers (mean [SD] SUVmax, 2.3 [0.5] vs 1.5 [0.3]; $P < .001$). Similar findings were observed for target to background ratio values (eTable 1 in Supplement 1).

The pattern of [^{68}Ga]FAPI-46 uptake differed depending on the underlying etiology of the heart failure. In patients with ischemic cardiomyopathy, increased [^{68}Ga]FAPI-46 uptake was consistently observed in areas of chronic myocardial infarction (Figure 2), with no increase in uptake observed in the remote myocardium or peri-infarct zone in comparison with healthy volunteers (mean [SD] SUVmax, 1.5 [0.4] vs 1.5 [0.3]; $P = .88$; and 1.6 [0.3] vs 1.5 [0.3]; $P = .59$, respectively) (eFigure 2 in Supplement 1).

A different pattern of uptake was observed in patients with nonischemic cardiomyopathy, with 15 patients (71%) displaying visually increased [^{68}Ga]FAPI-46 uptake within the basal septal region. Across the cohort as a whole, quantitative assessment confirmed increased [^{68}Ga]FAPI-46 in the septum compared with the rest of the left ventricle (mean [SD] SUVmax, 2.1 [0.5] in the septum vs 1.8 [0.5] in the rest of the left ventricle; $P = .001$) (Figure 3). In 9 patients (60%), this increased visual septal [^{68}Ga]FAPI-46 uptake colocalized with areas of noninfarct late gadolinium enhancement, while in the remaining 6 patients (40%), no underlying late gadolinium enhancement was

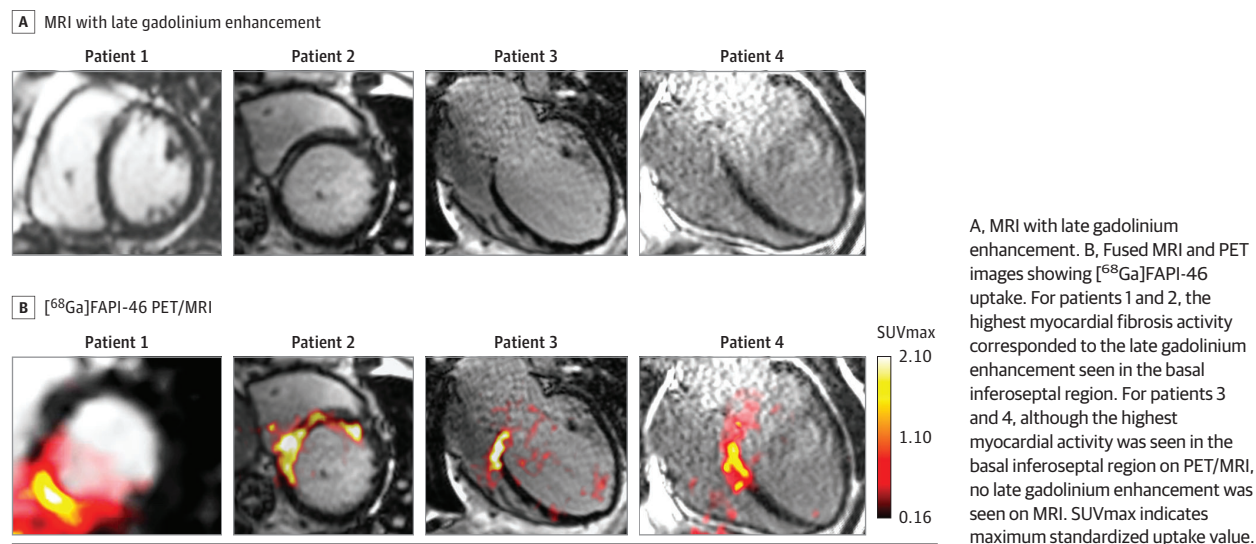
observed. After exclusion of the basal septum, patients continued to have diffusely increased [^{68}Ga]FAPI-46 uptake in the remaining segments of the left ventricle compared with healthy volunteers (mean [SD] SUVmax, 1.8 [0.5] vs 1.5 [0.3]; $P = .04$) (eFigure 2 in Supplement 1).

Patients With Myocardial Infarction

We compared myocardial [^{68}Ga]FAPI-46 uptake between the 21 patients with heart failure due to previous myocardial infarction (ischemic cardiomyopathy) and 20 patients with previous myocardial infarction but preserved LV function (eTable 2 in Supplement 1). Although there was a clear difference between the groups in terms of ejection fraction (mean [SD], 59% [8%] vs 40% [9%]; $P < .001$), there were no major differences observed in scar burden as measured by late gadolinium enhancement mass (mean [SD], 10.3 [8.0] vs 14.4 [21.0] g; $P = .11$). However, patients with impaired LV function and heart failure demonstrated higher myocardial [^{68}Ga]FAPI-46 uptake in their infarct zone compared with those with preserved LV ejection fraction and no heart failure (mean [SD] SUVmax, 3.2 [1.1] vs 2.5 [0.3]; $P = .03$) (eFigure 2 in Supplement 1).

Across all patients with previous myocardial infarction, there was no significant difference in myocardial [^{68}Ga]FAPI-46 uptake in patients with transmural ($n = 17$) vs non-transmural ($n = 25$) infarcts (mean [SD] SUVmax, 3.1 [1.3] vs 2.7 [1.1]; $P = .44$). Furthermore, myocardial [^{68}Ga]FAPI-46 uptake also did not differ based on the location of the infarct (anterior [$n = 20$], mean [SD] SUVmax, 3.1 [1.6]; inferior [$n = 17$], mean [SD] SUVmax, 2.6 [0.6]; lateral [$n = 4$], mean [SD] SUVmax, 2.6 [0.9]; $P = .39$).

Figure 3. Magnetic Resonance Imaging (MRI) and Gallium 68-Labeled Fibroblast Activation Protein Inhibitor 46 (^{68}Ga)FAPI-46 Positron Emission Tomography (PET)/MRI Examples From Patients With Nonischemic Cardiomyopathy



Predictors of Myocardial Fibroblast Activation

Across the cohort as a whole, myocardial ^{68}Ga FAPI-46 uptake showed a moderate positive correlation with late gadolinium enhancement mass ($r = 0.47$; $P = .003$) (eFigure 3 in Supplement 1), although no significant correlation with native T1 values was observed ($r = 0.24$; $P = .07$).

On univariable analysis within the heart failure cohort, increased high-sensitivity cardiac troponin T level, higher BNP level, and reduced LV ejection fraction were associated with increased myocardial ^{68}Ga FAPI-46 uptake (eTable 3 in Supplement 1). After stepwise multivariable regression including age, sex, ejection fraction, BNP level, and high-sensitivity cardiac troponin T level, only increased high-sensitivity cardiac troponin T level and reduced LV ejection fraction were independently associated with ^{68}Ga FAPI-46 uptake in heart failure (Akaike information criterion, -3.49 [$P = .05$] and -5.11 [$P = .13$], respectively). Furthermore, logistic regression using the entire study population (participants with and without heart failure) demonstrated that ^{68}Ga FAPI-46 uptake was higher in patients with reduced ejection fraction ($\rho = -0.54$; $P < .001$) (eFigure 4 in Supplement 1).

Repeat Imaging

Nineteen patients with heart failure (8 with ischemic cardiomyopathy and 11 with nonischemic cardiomyopathy) underwent repeat imaging 6 to 12 months after their baseline scan. There were no statistically significant differences in myocardial ^{68}Ga FAPI-46 uptake (mean [SD] SUVmax, 2.7 [0.7] at baseline vs 2.4 [0.7] at follow-up; $P = .06$) or LV ejection fraction (mean [SD], 41% [9%] at baseline vs 44% [9%] at follow-up; $P = .14$) observed over time. There was heterogeneity within the cohort with regard to change in ejection fraction, with 8 patients demonstrating a clear improvement of 5% or more in their ejection fraction at follow-up.

There were no correlations between baseline myocardial ^{68}Ga FAPI-46 uptake and change in high-sensitivity cardiac

troponin T level ($r = 0.13$; $P = .62$) or BNP level ($r = 0.04$; $P = .89$). However, higher baseline myocardial ^{68}Ga FAPI-46 uptake was associated with less improvement in ejection fraction with optimal medical therapy over time, suggesting that patients with higher myocardial fibroblast activation tended to show reduced functional recovery ($r = -0.52$; $P = .02$). Considered from the other perspective, the 8 patients who demonstrated clear improvement in their ejection fraction ($\geq 5\%$ improvement) at follow-up displayed lower myocardial ^{68}Ga FAPI-46 uptake at baseline than the 11 patients who did not demonstrate improvement in systolic function (mean [SD] SUVmax, 2.5 [0.5] vs 3.3 [0.5]; $P = .003$).

Discussion

In a prospective clinical study, we characterized myocardial fibroblast activation using ^{68}Ga FAPI-46 PET/MRI in patients with heart failure and reduced ejection fraction. These patients demonstrated increased myocardial fibroblast activation proportional to the extent of LV dysfunction and injury, with distinct distribution patterns in ischemic and nonischemic etiologies and an inverse association with subsequent changes in systolic function. These findings suggest that activated fibroblasts play an important pathophysiological role in heart failure progression and represent an important potential treatment target. Moreover, ^{68}Ga FAPI-46 PET/MRI may serve as a prognostic biomarker for risk stratification and as a measure of disease activity to assess novel therapies or treatment response.

To date, there have been only limited studies investigating myocardial ^{68}Ga FAPI-46 uptake in patients with heart failure. Liang et al¹⁹ recently reported fluorine 18-labeled FAPI imaging in 19 patients with dilated cardiomyopathy, with findings broadly consistent with ours. However, unlike their study, we found that volumetric assessment of myocardial fibro-

blast activation was not feasible. Although we have successfully used this approach in patients with acute myocardial infarction,¹⁷ [⁶⁸Ga]FAPI-46 uptake in nonischemic cardiomyopathy was too low and diffuse to allow any meaningful volumetric analysis using available software (QPET). Song et al²⁰ recently demonstrated increased [⁶⁸Ga]FAPI-46 uptake in a rat model of heart failure and in 7 patients with heart failure, and a retrospective oncology cohort study by Heckmann et al²¹ identified increased myocardial [⁶⁸Ga]FAPI-46 uptake in 4 patients with reduced ejection fraction. To our knowledge, we report here the largest prospective heart failure cohort to date, the first to compare patients with ischemic and nonischemic cardiomyopathy, and the first to perform serial imaging, confirming increased myocardial [⁶⁸Ga]FAPI-46 uptake and enabling subgroup analyses.

Myocardial infarction is the prototypical model of heart failure with myocyte necrosis, collagen deposition, and scar formation. Consistent with this, we observed the highest myocardial fibroblast activation in patients with ischemic cardiomyopathy. Interestingly, fibroblast activation in the remote myocardium and peri-infarct zone was comparable to that in healthy volunteers. Higher [⁶⁸Ga]FAPI-46 uptake was observed in patients with previous myocardial infarction and reduced ejection fraction compared with those with preserved ejection fraction, even though there was no major difference in the infarct size between the groups. These findings suggest that activated fibroblasts localize predominantly to the infarct zone and that greater fibroblast activation is associated with the development of heart failure, supporting a role for activated fibroblasts in adverse ventricular remodeling.

We observed a different pattern of myocardial fibroblast activation in patients with nonischemic cardiomyopathy compared with ischemic cardiomyopathy. Midwall fibrosis in the basal septum is present on late gadolinium enhancement in up to three-quarters of patients with nonischemic cardiomyopathy, and this pattern is associated with adverse cardiovascular outcomes, such as all-cause mortality, arrhythmic death, and hospitalization for heart failure.²²⁻²⁴ Consistent with these published cardiac MRI studies, we found that while diffuse increases in fibroblast activation could be observed across the LV in patients with nonischemic cardiomyopathy, the highest [⁶⁸Ga]FAPI-46 uptake was most frequently observed in the basal septum both with and without underlying late gadolinium enhancement. The reason for the development of septal fibrosis in nonischemic cardiomyopathy is not clearly understood but is often attributed to changes in LV mechanics and sites of increased wall stress. Future studies should investigate the progression of myocardial fibroblast activation and its association with adverse outcomes in patients with nonischemic cardiomyopathy.

Our findings have important clinical implications. [⁶⁸Ga]FAPI-46 imaging enables the first direct assessment of activated myocardial fibroblasts, key effector cells, and potential therapeutic target heart failure. Importantly, [⁶⁸Ga]FAPI-46 uptake provides information different from but complementary to that of cardiac MRI. While the latter provides information about established myocardial fibrosis, [⁶⁸Ga]FAPI-46 uptake identifies the cellular

activity associated with myocardial fibrosis and adverse remodeling, offering a marker of disease activity. Our study provides a foundation for future work exploring fibroblast activation across different subtypes of cardiomyopathies and heart failure phenotypes.

The ability to visualize activated fibroblasts and myocardial disease activity may facilitate targeted therapy development and enable monitoring of treatment response. Emerging heart failure therapies, like sodium-glucose cotransporter-2 inhibitors, glucagon-like peptide-1 receptor agonists, and pirfenidone, have demonstrated potential antifibrotic effects in preclinical models, but these have not been robustly confirmed in humans, largely due to the lack of noninvasive tools to quantify myocardial fibroblast activation.^{25,26} Additionally, promising experimental treatments, such as CAR T-cell therapy, may also benefit from [⁶⁸Ga]FAPI-46 imaging to evaluate cellular effects.^{27,28} Our longitudinal data offer some support to this potential role, showing stable myocardial uptake over time and an association between higher baseline uptake and less improvement in ejection fraction with medical therapy.

Limitations

We acknowledge that our study has limitations. First, although to our knowledge this is the largest prospective study to date, the study population size remains modest; large multicenter studies are needed to demonstrate the clinical utility and generalizability of our findings. Second, the radiotracer used here is not currently widely available for clinical use, although more widespread access to the tracer is anticipated in the near future. Third, our study was not specifically powered to assess the correlation between myocardial fibroblast activation and MRI parameters like late gadolinium enhancement mass in the different subgroups of patients, as larger studies are required to address this issue. Fourth, we have not assessed the time course of myocardial [⁶⁸Ga]FAPI-46 uptake beyond 6 months or its association with clinical events. Larger longitudinal cohort studies of patients with heart failure are required to investigate how fibroblast activation changes beyond 6 months and whether it can identify patients most likely to develop an adverse clinical disease course.

Conclusions

This prospective case-control human study demonstrates increased myocardial fibroblast activation in ischemic and nonischemic cardiomyopathies using [⁶⁸Ga]FAPI-46 PET/MRI. Fibroblast activation persisted over 6 months, was associated with changes in systolic function, and showed distinct etiology-specific patterns. These findings support an important pathophysiological role for activated fibroblasts in heart failure and identify them as a potential therapeutic target. [⁶⁸Ga]FAPI-46 PET/MRI enables in vivo assessment of fibroblast activation and may aid mechanistic understanding, risk stratification, and the development of targeted therapies in heart failure.

ARTICLE INFORMATION

Accepted for Publication: January 29, 2026.

Published Online: April 15, 2026.
doi:10.1001/jamacardio.2026.0661

Author Contributions: Dr Joshi had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Concept and design: Joshi, Barton, Williams, Baker, Dey, Slomka, Newby, Dweck.

Acquisition, analysis, or interpretation of data: Joshi, Barton, Loganath, Craig, Mahendran, Whittington, Ramtoola, Williams, van Beek, Fletcher, Clark, Waight, Dey, Slomka, Newby, Dweck.

Drafting of the manuscript: Joshi, Whittington, Dweck.

Critical review of the manuscript for important intellectual content: Joshi, Barton, Loganath, Craig, Mahendran, Ramtoola, Williams, van Beek, Baker, Fletcher, Clark, Waight, Dey, Slomka, Newby, Dweck.

Statistical analysis: Joshi.

Obtained funding: Joshi, Williams, Newby, Dweck.

Administrative, technical, or material support: Joshi, Loganath, Whittington, Ramtoola, Williams, van Beek, Fletcher, Clark, Waight, Dey, Slomka, Dweck. **Supervision:** van Beek, Dey, Newby, Dweck.

Conflict of Interest Disclosures: Dr Joshi reported receiving grants from the British Heart Foundation Clinical Research and AstraZeneca during the conduct of the study. Dr Williams reported receiving personal fees from Canon, Siemens Healthineers, Novartis, Bayer, and FEops outside the submitted work. Dr van Beek reported being an owner and founder of QCTIS Ltd and receiving personal fees from AstraZeneca, DeepHealth/Aidence, and Lunit outside the submitted work. Dr Dey reported receiving software royalties from Cedars-Sinai Medical Center and grant support from the National Heart, Lung, and Blood Institute, National Institutes of Health outside the submitted work. Dr Slomka reported having equity interest in APQ Health, receiving royalties from Cedars-Sinai Medical Center, and receiving personal fees from Synektik and Novo Nordisk outside the submitted work. No other disclosures were reported.

Funding/Support: This work was supported by a clinical research training fellowship from the British Heart Foundation (FS/CRTF/20/24087; Dr Joshi) and an investigator-initiated award from AstraZeneca (ESR-19-20118). Dr Joshi is supported by the Chief Scientist Office (PCL/24/03). Dr Craig is supported by the Medical Research Council (MR/Y009932/1). Drs Williams, Newby, and Dweck are supported by the British Heart Foundation (CH/09/002, FS/ICRF/20/26002, FS/SCRF/21/32010, RG/23/F/22/110093, RE/24/130012), Medical Research Council (MR/Y008944/1, MR/Z504658/1, G0701127), and the joint Medical Research Council and British Heart Foundation Centre of Research Excellence in Advanced Cardiac Therapies (REACT; SI/F/24/21170016). The Edinburgh Imaging Facility and Edinburgh Clinical Research Facility are supported by the National Health Service Research Scotland through the National Health Service Lothian Health Board.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or

approval of the manuscript; or decision to submit the manuscript for publication.

Data Sharing Statement: See Supplement 2.

REFERENCES

- Taylor CJ, Ordóñez-Mena JM, Roalfe AK, et al. Trends in survival after a diagnosis of heart failure in the United Kingdom 2000-2017: population based cohort study. *BMJ*. 2019;364(February):1223. doi:10.1136/bmj.l223
- Konstam MA, Kramer DG, Patel AR, Maron MS, Udelsion JE. Left ventricular remodeling in heart failure: current concepts in clinical significance and assessment. *JACC Cardiovasc Imaging*. 2011;4(1):98-108. doi:10.1016/j.jcmg.2010.10.008
- Thackeray JT, Hupe HC, Wang Y, et al. Myocardial inflammation predicts remodeling and neuroinflammation after myocardial infarction. *J Am Coll Cardiol*. 2018;71(3):263-275. doi:10.1016/j.jacc.2017.11.024
- Libby P, Theroux P. Pathophysiology of coronary artery disease. *Circulation*. 2005;111(25):3481-3488. doi:10.1161/CIRCULATIONAHA.105.537878
- Stary HC, Chandler AB, Glagov S, et al. A definition of initial, fatty streak, and intermediate lesions of atherosclerosis: a report from the Committee on Vascular Lesions of the Council on Arteriosclerosis, American Heart Association. *Circulation*. 1994;89(5):2462-2478. doi:10.1161/01.CIR.89.5.2462
- Loganath K, Craig N, Barton A, et al. Cardiovascular positron emission tomography imaging of fibroblast activation: a review of the current literature. *J Nucl Cardiol*. 2024;47:102106. doi:10.1016/j.nuclcard.2024.102106
- Barton AK, Tzolos E, Bing R, et al. Emerging molecular imaging targets and tools for myocardial fibrosis detection. *Eur Heart J Cardiovasc Imaging*. 2023;24(3):261-275. doi:10.1093/ehjci/jeac242
- Zhang M, Quan W, Zhu T, et al. [⁶⁸Ga]Ga-DOTA-FAPI-04 PET/MR in patients with acute myocardial infarction: potential role of predicting left ventricular remodeling. *Eur J Nucl Med Mol Imaging*. 2023;50(3):839-848. doi:10.1007/s00259-022-06015-0
- Diekmann J, Koenig T, Thackeray JT, et al. Cardiac fibroblast activation in patients early after acute myocardial infarction: integration with MR tissue characterization and subsequent functional outcome. *J Nucl Med*. 2022;63(9):1415-1423. doi:10.2967/jnumed.121.263555
- Giesel FL, Adebeg S, Syed M, et al. FAPI-74 PET/CT using either ¹⁸F-ALF or cold-kit ⁶⁸Ga labeling: biodistribution, radiation dosimetry, and tumor delineation in lung cancer patients. *J Nucl Med*. 2021;62(2):201-207. doi:10.2967/jnumed.120.245084
- Kratochwil C, Flechsig P, Lindner T, et al. ⁶⁸Ga-FAPI PET/CT: tracer uptake in 28 different kinds of cancer. *J Nucl Med*. 2019;60(6):801-805. doi:10.2967/jnumed.119.227967
- Totzeck M, Siebermair J, Rassaf T, Rischpler C. Cardiac fibroblast activation detected by positron emission tomography/computed tomography as a possible sign of cardiotoxicity. *Eur Heart J*. 2020;41(9):1060. doi:10.1093/eurheartj/ehz736
- Maleki S, Esmaili Z, Seighali N, et al. Cardiac adverse events after chimeric antigen receptor (CAR) T cell therapies: an updated systematic review and meta-analysis. *Cardiooncology*. 2024;10(1):52. doi:10.1186/s40959-024-00252-y
- McDonagh TA, Metra M, Adamo M, et al; ESC Scientific Document Group. 2023 Focused update of the 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2023;44(37):3627-3639. doi:10.1093/eurheartj/ehad195
- Arbello E, Protonotarios A, Gimeno JR, et al; ESC Scientific Document Group. 2023 ESC guidelines for the management of cardiomyopathies. *Eur Heart J*. 2023;44(37):3503-3626. doi:10.1093/eurheartj/ehad194
- Robson PM, Dweck MR, Trivieri MG, et al. Coronary artery PET/MR imaging: feasibility, limitations, and solutions. *JACC Cardiovasc Imaging*. 2017;10(pt A):1103-1112. doi:10.1016/j.jcmg.2016.09.029
- Barton AK, Craig NJ, Loganath K, et al. Myocardial fibroblast activation after acute myocardial infarction: a positron emission tomography and magnetic resonance study. *J Am Coll Cardiol*. 2025;85(6):578-591. doi:10.1016/j.jacc.2024.10.103
- McDonagh TA, Metra M, Adamo M, et al; ESC Scientific Document Group. 2021 ESC guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42(36):3599-3726. doi:10.1093/eurheartj/ehab368
- Liang S, Hou P, Wang X, et al. Comparison of ¹⁸F-FAPI-42 PET for detecting cardiac fibroblast activation in dilated cardiomyopathy with histopathology and CMR. *JACC Cardiovasc Imaging*. 2025;18(9):997-1009. doi:10.1016/j.jcmg.2025.05.021
- Song W, Zhang X, He S, et al. ⁶⁸Ga-FAPI PET visualize heart failure: from mechanism to clinic. *Eur J Nucl Med Mol Imaging*. 2023;50(2):475-485. doi:10.1007/s00259-022-05994-4
- Heckmann MB, Reinhardt F, Finke D, et al. Relationship between cardiac fibroblast activation protein activity by positron emission tomography and cardiovascular disease. *Circ Cardiovasc Imaging*. 2020;13(9):e010628. doi:10.1161/CIRCIMAGING.120.010628
- Ballantyne B, Vandenberk B, Dykstra S, et al. Patients with non-ischemic cardiomyopathy and mid-wall striae have similar outcomes as patients with ischemic cardiomyopathy: a propensity-matched analysis. *Can J Cardiol*. 2022;38(10)(suppl 2):S179. doi:10.1016/j.cjca.2022.08.139
- Chevalier C, Kremer K, Cavus E, et al. CMR feature tracking in patients with dilated cardiomyopathy: patterns of myocardial strain and focal fibrosis. *Open Heart*. 2022;9(2):e002013. doi:10.1136/openhrt-2022-002013
- Csecs I, Pashakhanloo F, Paskavitz A, et al. Association between left ventricular mechanical deformation and myocardial fibrosis in nonischemic cardiomyopathy. *J Am Heart Assoc*. 2020;9(19):e016797. doi:10.1161/JAHA.120.016797
- Shakour N, Karami S, Iranshahi M, Butler AE, Sahebkar A. Antifibrotic effects of sodium-glucose cotransporter-2 inhibitors: a comprehensive review. *Diabetes Metab Syndr*. 2024;18(1):102934. doi:10.1016/j.dsx.2023.102934
- Graziani F, Lillo R, Crea F. Rationale for the use of pirfenidone in heart failure with preserved ejection fraction. *Front Cardiovasc Med*. 2021;8:678530. doi:10.3389/fcvm.2021.678530
- Taherkhani S, Honardoost M, Dokhani N, Janzadeh A. Revolutionizing cardiac fibrosis treatment: the potential of personalized CAR T-cell therapy. *Cardiooncology*. 2025;11(1):76. doi:10.1186/s40959-025-00367-w
- Aghajanian H, Kimura T, Rurik JG, et al. Targeting cardiac fibrosis with engineered T cells. *Nature*. 2019;573(7774):430-433. doi:10.1038/s41586-019-1546-z